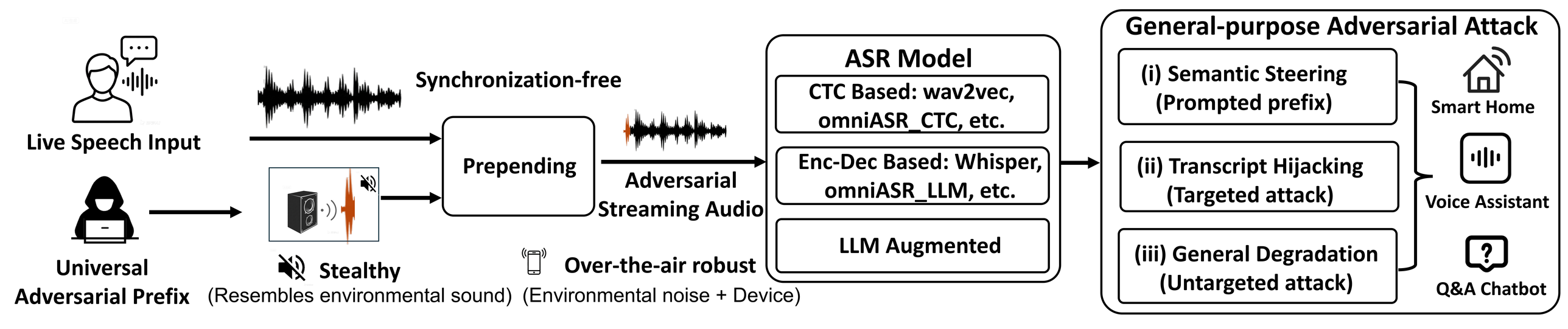
Live Speech Input
Synchronization-free
Universal
Adversarial Prefix
Stealthy
(Resembles environmental sound)
Prepending
Adversarial
Streaming Audio
Over-the-air robust
(Environmental noise + Device)
ASR Model
CTC Based: wav2vec,
omniASR_CTC, etc.
Enc-Dec Based: Whisper,
omniASR_LLM, etc.
LLM Augmented
General-purpose Adversarial Attack
(i) Semantic Steering
(Prompted prefix)
(ii) Transcript Hijacking
(Targeted attack)
(iii) General Degradation
(Untargeted attack)
Smart Home
Voice Assistant
Q&A Chatbot